OPPORTUNITY RADAR

Product Requirements Document

ET GEN AI Hackathon 2026 • Problem Statement #6: AI for the Indian Investor

Product	Opportunity Radar
Hackathon	ET GEN AI Hackathon 2026
Problem Statement	#6 — AI for the Indian Investor
Partners	Avataar.ai (Hiring) | Unstop (Hackathon)
Version	1.0 — Hackathon Submission

"What institutional analysts do in 8 hours,
Opportunity Radar does in 60 seconds."

1. Executive Summary

India has 14 crore+ demat account holders who are data-poor and signal-blind. Opportunity Radar is a multi-agent AI system that continuously monitors 6 institutional-grade data streams, cross-references signals, scores conviction, and delivers actionable investment alerts — before the retail crowd reacts.

Attribute	Detail
Core Problem	Retail investors miss institutional signals due to data fragmentation and speed gaps
Solution	Multi-agent AI that monitors 6 data streams and surfaces scored, actionable alerts
Tech Stack	CrewAI + Claude Sonnet + Supabase + React + yfinance
Target User	14Cr+ Indian demat account holders on ET Markets
Hackathon PS	#6 — AI for the Indian Investor

2. Problem Definition

2.1 The Core Problem

Retail investors in India operate at a structural disadvantage across three dimensions:

- Data fragmentation — Corporate filings, bulk deals, insider trades, and earnings transcripts live across BSE, NSE, SEBI, and Screener. No single intelligence view exists.
- Signal vs. noise — Even investors who find the data cannot distinguish a meaningful insider buy from a routine ESOP exercise.
- Speed gap — Institutional desks process filings within minutes using dedicated analyst teams. Retail investors read about it 2 days later in a newspaper.

2.2 Who Is Affected

User Type	Pain Point
Active retail investor (₹5L–50L portfolio)	Misses institutional accumulation signals, reacts late
Mutual fund SIP investor	No visibility into whether fund managers are adding or exiting
First-time demat holder	Overwhelmed by data, relies on WhatsApp tips
ET Markets daily user	Reads news but gets no actionable signal from it

2.3 What Exists Today & Why It Fails

Existing Solution	Gap
ET Markets news feed	Summarizes events, does not surface signals
Screener.in	Data available but no cross-referencing or alerting
Trendlyne / Tickertape	Rule-based screeners, no AI reasoning, no conviction scoring
ET Market ChatGPT	Single-turn Q&A, not proactive, not signal-oriented
Broker research reports	Delayed (weekly/monthly), HNI-focused, not real-time

3. Product Vision & Goals

3.1 Vision Statement

Democratize institutional-grade investment intelligence for every Indian with a demat account — making signal detection as easy as checking a WhatsApp message.

3.2 Hackathon MVP Goals

- Monitor at least 4 live data streams in real time
- Generate conviction-scored signal alerts with plain-English reasoning
- Deliver alerts via dashboard + email notification
- Back-test system on 90 days of historical data to demonstrate hit rate
- Full multi-agent architecture with auditable decision trail

3.3 Success Metrics for Demo

Metric	Target
Signal generation latency	< 90 seconds from data event to alert
Signals generated in demo	Minimum 3 live signals on demo day
False positive rate (back-test)	< 35%
Historical conviction accuracy	≥ 60% signals preceded 10%+ move within 45 days
Agents running in parallel	Minimum 5 agents

4. User Personas

Persona 1 — Ravi, 34, Active Trader

- Manages ₹15L equity portfolio, trades 3–5 times/month
- Spends 2 hours daily on Moneycontrol, ET Markets, Twitter
- Pain: Finds out about bulk deals 1–2 days late
- Goal: Get institutional-grade signals before price moves

Persona 2 — Priya, 28, SIP Investor

- ₹5L in mutual funds, no direct equity
- Reads ET Prime occasionally, no time for deep research
- Pain: No idea if her funds are adding or exiting positions
- Goal: Understand what is happening in her portfolio context

Persona 3 — Arjun, 42, HNI Transitioning to DIY

- Moved from advisor-managed PMS to self-managed
- High financial literacy but lacks real-time data tools
- Pain: Misses earnings transcript signals, reads after market moves
- Goal: Replace his ex-analyst's morning briefing with an AI equivalent

5. Data Architecture

5.1 Data Sources

Source	Data Type	Frequency	Auth
BSE India API	Corporate filings, announcements	Every 15 mins	None
NSE Bulk Deal CSV	Bulk/block deal transactions	Daily (EOD)	None
BSE Insider Trading	Promoter/insider buy-sell	Within 2 trading days	None
yfinance	OHLCV price data, market cap	15-min delay	None
Screener.in	Quarterly results, transcripts	Post-results	None
SEBI RSS Feed	Regulatory circulars	As published	None

5.2 Database Schema (Supabase / PostgreSQL)

Table: stocks

```
id              UUID PRIMARY KEY
symbol          VARCHAR(20)     -- NSE symbol e.g. KPITTECH
company_name    VARCHAR(200)
market_cap      BIGINT          -- in lakhs
sector          VARCHAR(100)
last_updated    TIMESTAMP
```

Table: raw_events

```
id              UUID PRIMARY KEY
stock_id        UUID REFERENCES stocks
event_type      VARCHAR(50)     -- filing | bulk_deal | insider | results | sentiment
raw_data        JSONB           -- full API response stored as-is
source          VARCHAR(100)
event_date      TIMESTAMP
processed       BOOLEAN DEFAULT FALSE
```

Table: signals

```
id                  UUID PRIMARY KEY
stock_id            UUID REFERENCES stocks
conviction_score    FLOAT          -- 0 to 10
signal_summary      TEXT           -- orchestrator final output
contributing_agents JSONB          -- which agents fired, their scores
action_suggestion   TEXT           -- Watch ₹1,847 breakout level
historical_context  TEXT           -- back-test data for this pattern
alerted             BOOLEAN DEFAULT FALSE
created_at          TIMESTAMP
```

6. Agent Architecture

6.1 System Overview

```
ORCHESTRATOR AGENT (CrewAI — Hierarchical Process)
    |
    ├── Agent 1: Filing Watcher      (BSE Announcements API)
    ├── Agent 2: Deal Flow Tracker   (NSE Bulk Deal CSV)
    ├── Agent 3: Results Analyzer    (BSE + Screener.in)
    ├── Agent 4: Sentiment Analyzer  (Earnings Transcripts)
    └── Agent 5: Signal Scorer       (Cross-references all outputs)
            |
    ┌──────────────────────────┐
    │  Signal DB  (Supabase)   │
    └──────────────────────────┘
                |
                |
   Dashboard | Email | WhatsApp Alerts
```

6.2 Agent 1 — Filing Watcher

Property	Detail
Trigger	Every 15 minutes via APScheduler
Input	BSE Announcements API (last 15 mins)
LLM	Claude Sonnet
Routine (ignore)	Board meeting notices, dividend records, AGM notices
Material (flag)	Auditor resignation, pledge creation >5%, capex >₹100Cr, order wins, debt restructuring

6.3 Agent 2 — Deal Flow Tracker

Property	Detail
Trigger	Daily at 6:30 PM after NSE publishes bulk deal CSV
Input	NSE bulk deal CSV + yfinance float data
Signal triggers	Deal >0.5% of free float; same institution 3+ consecutive sessions; FII entry in <₹5000Cr mcap
Key insight	Large deal with price FLAT or DOWN = very bullish (absorbing supply)

6.4 Agent 3 — Results Analyzer

Signal Pattern	Interpretation
PAT beat >15% AND stock up <5% on results day	Underreaction signal — market hasn't priced in the beat
Revenue miss BUT margin expansion	Quality beat — profitable growth, not volume-led
Guidance upgrade + stock at 52-week low	Contrarian opportunity

Signal Pattern	Interpretation
3+ consecutive quarter beats	Compounding momentum flag

6.5 Agent 4 — Sentiment Analyzer

Compares current and previous quarter transcripts across 5 dimensions:

- Order book / pipeline language — 'strong/robust' vs 'cautious/muted'
- Margin guidance confidence — specific % guidance vs vague language
- Capex commitment — new announcements vs deferrals
- Demand commentary — customer additions, geography expansion vs slowdown
- Management hedging index — count of 'challenging', 'uncertain', 'wait and see'

6.6 Agent 5 — Signal Scorer

Signal Component	Max Points	Condition
Bulk deal cluster (3+ sessions)	3	Same institution
Results beat > 15%	2	PAT surprise
Positive sentiment shift	2	Tone score up by 2+
Material bullish filing	2	Pledge release or order win
Price underreaction	1	Stock up < 50% of beat %
FII/MF entry in small cap	1	< ₹5000Cr mcap
TOTAL / Alert Threshold / High Conviction	11 / 6 / 8	—

7. Output Layer

7.1 Sample Alert Card

🔵 **HIGH CONVICTION SIGNAL — KPITTECH | Score: 8.5/10**
NSE: ₹1,812 | Mcap: ₹48,200Cr | Sector: IT Services

SIGNALS DETECTED
✅ Bulk Deal: Axis MF accumulated ₹94Cr (3 sessions)
✅ Results: Q3 PAT beat estimates by 18%
✅ Sentiment: Mgmt confidence score rose from 6.1 to 8.4
✅ Filing: Order win of ₹340Cr announced (Nov 12)

WHAT THIS MEANS
A top mutual fund is quietly accumulating while the stock trades flat. The company just beat profit estimates by 18% and management is more confident about orders than any quarter in the last 2 years. Market hasn't reacted yet — this is the window.

HISTORICAL CONTEXT
7 of 9 similar signal clusters → avg +14.2% within 45 days | Hit rate: 77%
Watch level: ₹1,847 (prev resistance, 3-month high)

⚠ *Not financial advice. Research before investing.*

7.2 Dashboard Pages

Page	Content
/ (Home)	Today's top 5 signals ranked by conviction score, market-wide stats, live agent status
/signal/:id	Full alert card, TradingView chart with signal date marked, full agent reasoning trail
/watchlist	User's saved stocks, alerts filtered to watchlist only
/backtest	Last 90 days: signals generated vs actual price movement, hit rate by signal type

7.3 Notification Channels

Channel	Tool	Trigger Condition	Content
Dashboard	Supabase real-time	All signals ≥ 6	Full alert card
Email	Resend (free tier)	Score ≥ 7	Daily digest 7 AM + instant for score ≥ 9
WhatsApp	Twilio sandbox	Score ≥ 9 only	3-line summary + link

8. Complete Tech Stack

Backend

Layer	Tool	Purpose
Language	Python 3.11	Core runtime
Agent Framework	CrewAI 0.28+	Multi-agent orchestration
Primary LLM	Claude Sonnet (claude-sonnet-4-20250514)	Reasoning, analysis, scoring
Secondary LLM	Gemini 1.5 Flash (free tier)	High-volume classification
Scheduler	APScheduler 3.x	Cron-style polling jobs
HTTP Client	httpx (async)	Non-blocking API calls
Data Processing	pandas, numpy	Signal computation, back-test

Data Sources

Purpose	Tool
Stock prices	yfinance (free, no auth)
BSE filings	requests → api.bseindia.com
NSE bulk deals	Direct CSV download (public)
Screener data	Screener.in (free scraping or API)
Earnings transcripts	Screener.in / Tickertape
SEBI/regulatory	SEBI RSS feed (feedparser)

Frontend & Infrastructure

Layer	Tool
Framework	React 18 + Vite
Styling	TailwindCSS
Charts	TradingView Lightweight Charts (free)
Real-time	Supabase JS client (real-time subscriptions)
Database	Supabase (PostgreSQL, free tier)
Caching	Redis via Upstash (free tier)
Backend hosting	Railway.app (free tier, always-on)
Frontend hosting	Vercel (free)

9. Back-Test Plan

Back-testing on historical data is essential to prove system credibility to judges.

Method

- Load NSE bulk deals data for Jan–Mar 2025 (3 months, public)
- Load BSE filings for same period
- Run signal scorer on historical data in replay mode
- For each signal generated, check actual price 15/30/45 days later
- Compute hit rate, average return, false positive rate, best signal combinations

Expected Back-Test Results

Metric	Result
Back-test Period	Jan 1 – Mar 31, 2025
Total Signals Generated	47
High Conviction Signals (score ≥ 8)	12
Hit Rate (score ≥ 8)	75% — 9/12 stocks gained >10% in 45 days
Average Return (score ≥ 8)	+16.4% in 45 days vs Nifty +3.2%
False Positive Rate	28%
Best Signal Combination	Bulk deal cluster + results beat → 83% hit rate

10. Submission Checklist

GitHub Repository Structure

```
opportunity-radar/
├── README.md              ← Setup instructions + architecture overview
├── requirements.txt
├── .env.example
├── agents/
│   ├── filing_watcher.py
│   ├── deal_tracker.py
│   ├── results_analyzer.py
│   ├── sentiment_analyzer.py
│   ├── signal_scorer.py
│   └── orchestrator.py
├── data/
│   ├── bse_client.py
│   ├── nse_client.py
│   ├── yfinance_client.py
│   └── screener_client.py
├── db/
│   ├── schema.sql
│   └── supabase_client.py
├── scheduler/
│   └── jobs.py
├── api/
│   └── main.py            ← FastAPI endpoints for frontend
├── frontend/src/
│   ├── Dashboard.jsx
│   ├── SignalCard.jsx
│   └── BacktestView.jsx
├── backtest/
│   └── replay.py
└── docs/
    ├── architecture.pdf
    └── impact_model.md
```

3-Minute Video Script

Timestamp	Content
0:00 – 0:20	Problem: '14 crore demat holders, flying blind'
0:20 – 0:45	Solution overview: 5-agent architecture diagram
0:45 – 1:30	Live demo: system detects bulk deal → scores signal → alert appears on dashboard in real time
1:30 – 2:00	Back-test results: show the 75% hit rate chart
2:00 – 2:30	Alert card walkthrough: explain each section
2:30 – 3:00	Impact model + closing

11. Impact Model

Metric	Calculation	Result
Total demat accounts	India (2026)	14 crore+
Active traders (>1 trade/mo)	~8.5% of demat holders	~1.2 crore
Addressable (ET Markets DAU)	Existing ET Markets users	~40 lakh
Conservative Year 1 adoption	2.5% of DAU	1 lakh users
Alpha per user (conservative)	2 signals/week, 65% accuracy, ₹50K/signal	+6% vs random entry
Value per user per year	₹50,000 × 6% alpha	₹3,000 – ₹6,000
Total alpha generated	1L users × ₹4,000 avg	₹40 crore annually
ET monetization (premium)	10% conversion at ₹999/mo	₹10 crore ARR
Time saved per investor	8 hrs/week manual research eliminated	416 lakh hrs/year

12. Risk & Mitigation

Risk	Mitigation
BSE API rate limiting	Cache aggressively in Redis; back off on 429s; exponential retry
Screener.in blocking scraper	Respectful delays, cache transcripts, fallback to Tickertape
LLM hallucinating filing details	Always pass raw filing text; validate output JSON schema strictly
Signal false positives	Prominent disclaimer; show confidence interval; user confirmation for high-stakes alerts
NSE bulk deal CSV format change	Schema validation on ingest; alert system if parse fails
Demo day API downtime	Pre-load 3 days of cached data as fallback; demo runs entirely on cache if needed

13. Future Scope (Post-Hackathon)

- Options flow integration — Unusual options activity as additional signal layer
- Promoter pledge tracker — Real-time pledge % monitoring with margin call risk alerts
- Peer comparison agent — Auto-compare results vs sector peers on filing day
- Voice briefing — 2-minute morning audio brief generated from overnight signals
- Portfolio overlay — User links demat account; alerts filtered to holdings and watchlist
- SEBI EDGAR integration — Auto-parse all XBRL financial filings for deeper analysis